

```
In [2]: import numpy as np
        from scipy import stats
        # Sample data
        marks = np.array([72, 68, 75, 70, 74, 69, 71, 73, 70, 72])
        # Hypothesized mean
        mu_0 = 70
        # One-sample t-test
        t_stat, p_value = stats.ttest_1samp(marks, mu_0)
        print(f"T-statistic: {t_stat:.3f}")
        print(f"P-value: {p_value:.4f}")
        alpha = 0.05
        if p_value < alpha:
            print("Reject Null Hypothesis → Mean is significantly different from 70.")
        else:
            print("Fail to Reject Null Hypothesis → No significant difference.")
```

T-statistic: 1.993

P-value: 0.0774

Fail to Reject Null Hypothesis → No significant difference.

```
In [3]: import numpy as np
        from math import sqrt
        from scipy.stats import norm
        # Given data
        x_bar = 51.2 # sample mean
        mu_0 = 50 # population mean
        sigma = 3 # population standard deviation
        n = 36 # sample size
        # Calculate Z-statistic
        z_stat = (x_bar - mu_0) / (sigma / sqrt(n))
        # Two-tailed p-value
        p_value = 2 * (1 - norm.cdf(abs(z_stat)))
        print(f"Z-statistic: {z_stat:.3f}")
        print(f"P-value: {p_value:.4f}")
        alpha = 0.05
        if p_value < alpha:
            print("Reject Null Hypothesis → Mean is significantly different from 50 g.")
        else:
            print("Fail to Reject Null Hypothesis → No significant difference.")
```

Z-statistic: 2.400

P-value: 0.0164

Reject Null Hypothesis → Mean is significantly different from 50 g.

```
In [4]: import numpy as np
        from scipy import stats

        # Data
        A = [20, 22, 23]
        B = [19, 20, 18]
        C = [25, 27, 26]

        # Perform one-way ANOVA
        f_stat, p_value = stats.f_oneway(A, B, C)
```

```
print(f"F-statistic: {f_stat:.3f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject Null Hypothesis → Means are significantly different.")
else:
    print("Fail to Reject Null Hypothesis → No significant difference.")
```

F-statistic: 25.923

P-value: 0.0011

Reject Null Hypothesis → Means are significantly different.

In []: